

Design Proposal

Project 20: Wi-Fi Password Cracking Attack

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Section / Group: B1 — Group 3

Updated per supervisor: multiple attacks on one setup; attacks run on captured pcap files (public sample pcaps allowed).

1 Overview of the Project Idea

Setup (one): Own offline tool that reads 802.11 pcaps, parses EAPOL, and recovers WPA2-PSK passphrases. Same host, same code base, same workflow.

Multiple attacks on that setup:

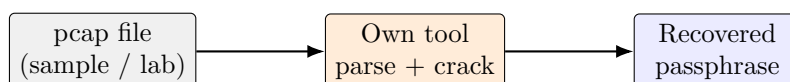
1. **Handshake MIC dictionary** — verify candidates against 4-way handshake MIC (He & Mitchell, 2004/2005).
2. **PMKID dictionary** — verify candidates against PMKID (CERT-EU SA2018-019, 2018).
3. **Mask / limited brute** — same verifiers, but candidates from a pattern (e.g. digits/short masks) instead of a wordlist.

Input = **pcap** (public demo/sample captures from the Internet, and/or our own lab capture). Cracking code is **ours** (Python/C; `hashlib/hmac`). No aircrack-ng / Hashcat.

Defense: Strong PSK + **WPA3-SAE-only** (RFC 7664; Lancrenon & Škrobot 2015). Avoid WPA2 transition mode (Dragonblood, 2020).

2 Network Topology, Components and Timing Diagram

2.1 Topology (one setup)

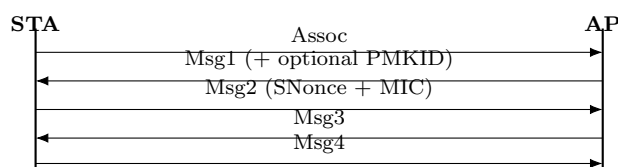


Optional lab AP/STA only to *produce* a pcap. Demo attacks run **on pcaps**, not against live foreign networks.

2.2 Components

Part	Role
pcap input	Public sample WPA2 handshake/PMKID captures; optional own lab capture.
Own tool	One pipeline: parse → Attack 1 / 2 / 3 → result.
Wordlist / mask	Candidates for dictionary vs mask modes.
Defense notes	WPA3-SAE-only + strong passphrase checklist.

2.3 Timing — handshake (in pcap)



2.4 Timing — attacks (offline on pcap)

Load pcap → extract EAPOL fields → run Attack 1, then 2, then 3 on the same capture (when fields exist) → report passphrase or “not found.”

3 Packet and Frame Structure

Layer	Fields
Radiotap / 802.11	Enough to locate data/EAPOL frames; MACs / BSSID.
LLC/SNAP	EtherType 0x888E.
EAPOL-Key	ANonce, SNonce, MIC, Key Data (PMKID / RSN IE).

PMK = PBKDF2-HMAC-SHA1(passphrase, SSID, 4096, 32)

Attack 1: PTK from PMK + nonces + MACs; match MIC.

Attack 2: PMKID = HMAC-SHA1-128(PMK, “PMK Name” || MAC_{AP} || MAC_{STA}).

Attack 3: same checks; different candidate generator (mask).

4 Implementation Plan

1. Collect sample WPA2 pcaps (public demo captures) + small wordlist/mask.
2. Build pcap reader + EAPOL parser (own code).
3. Implement Attack 1 (MIC), Attack 2 (PMKID), Attack 3 (mask).
4. Demo all three on the same setup/pcap set.
5. Defense section: strong PSK + WPA3-SAE-only; Dragonblood caveat.

Modules: pcap/, parse/, crypto/, attack_mic/, attack_pmkid/, attack_mask/, defend/.

5 Expected Outcome of a Successful Attack

- Attack 1 recovers passphrase from a handshake pcap when it is in the list.
- Attack 2 recovers passphrase from a PMKID-bearing pcap when present.
- Attack 3 recovers a short/pattern passphrase via mask within demo time.
- Wrong candidates fail; missing fields → that attack reports N/A.
- Defense contrast: WPA3-SAE is not crackable by these WPA2 offline methods.

6 Defense Idea

1. Long, non-dictionary passphrase.
2. Prefer **WPA3-SAE-only** (disable WPA2 transition).
3. Optional helper: warn if capture/settings imply WPA2-PSK only.

Dragonblood (2020): keep SAE implementations updated; avoid compatibility mode that re-enables WPA2-style offline cracking paths.